Article published online: 2024-04-23

Exploring the Limits of Artificial Intelligence for Referencing Scientific Articles

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Am J Perinatol 2024;41:2072–2081.

Abstract

Objective To evaluate the reliability of three artificial intelligence (AI) chatbots (ChatGPT, Google Bard, and Chatsonic) in generating accurate references from existing obstetric literature.

Study Design Between mid-March and late April 2023, ChatGPT, Google Bard, and Chatsonic were prompted to provide references for specific obstetrical randomized controlled trials (RCTs) published in 2020. RCTs were considered for inclusion if they were mentioned in a previous article that primarily evaluated RCTs published by the top medical and obstetrics and gynecology journals with the highest impact factors in 2020 as well as RCTs published in a new journal focused on publishing obstetric RCTs. The selection of the three AI models was based on their popularity, performance in natural language processing, and public availability. Data collection involved prompting the AI chatbots to provide references according to a standardized protocol. The primary evaluation metric was the accuracy of each AI model in correctly citing references, including authors, publication title, journal name, and digital object identifier (DOI). Statistical analysis was performed using a permutation test to compare the performance of the AI models.

Results Among the 44 RCTs analyzed, Google Bard demonstrated the highest accuracy, correctly citing 13.6% of the requested RCTs, whereas ChatGPT and Chatsonic exhibited lower accuracy rates of 2.4 and 0%, respectively. Google Bard often substantially outperformed Chatsonic and ChatGPT in correctly citing the studied reference components. The majority of references from all AI models studied were noted to provide DOIs for unrelated studies or DOIs that do not exist.

Conclusion To ensure the reliability of scientific information being disseminated, authors must exercise caution when utilizing AI for scientific writing and literature search. However, despite their limitations, collaborative partnerships between AI systems and researchers have the potential to drive synergistic advancements, leading to improved patient care and outcomes.

Keywords
- ChatGPT
- Chatsonic
- Google Bard
- artificial intelligence
- references

received
November 13, 2023
accepted
March 24, 2024
article published online
April 23, 2024

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Thieme Medical Publishers, Inc.,
333 Seventh Avenue, 18th Floor,
New York, NY 10001, USA

DOI https://doi.org/10.1055/s-0044-1786033.
ISSN 0735-1631.

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Key Points

- AI chatbots often cite scientific articles incorrectly.
- AI chatbots can create false references.
- Responsible AI use in research is vital.

Evidence-based medicine is a crucial component in the decision-making process for health care professionals. It relies heavily on high-quality scientific research, which can only be achieved by adhering to rigorous scientific standards. Accurate and comprehensive referencing is paramount for enhancing credibility and enabling verification of information. It is the responsibility of authors to prepare precise and relevant references that ensure acceptance within the scientific community and a seamless flow of information. Citation errors can have severe consequences, such as misinterpretation of research, inaccurate data analysis, and ethical violations, which can ultimately impact research validity and hinder scientific progress. The prevalence of citation errors in scientific literature has been estimated in several studies, with reported rates ranging from 4.1 to 54%.[1–4] More recently, a study conducted in 2017 estimated the rate of citation errors to be 14.5%.[5]

With the rise of artificial intelligence (AI) in various fields, including scientific research, AI chatbots like ChatGPT (OpenAI, San Francisco, CA) have become popular tools for assisting authors in writing and searching the scientific literature.[6–8] AI chatbots are designed to use a combination of natural language processing and machine learning algorithms to generate summaries based on patterns and associations found in their vast database. However, concerns have been raised regarding the accuracy of these tools and their potential to produce "hallucinations" or incorrect information.[9–15]

As AI continues to evolve, it is important for clinicians, researchers, and authors to consider the limitations of these tools when using them to answer clinical questions and for scientific research. These limitations include the absence of domain-specific knowledge, inability to access certain databases or resources, insufficient quality control measures, and a lack of reproducibility. The implications of AI chatbots' limitations on medical research and patient care are significant, as inaccurate, or incomplete literature searches can lead to biased conclusions and potentially ineffective or harmful treatments.

This cross-sectional study aims to investigate the accuracy of three popular AI chatbots in citing scientific articles to further explore the implications of using AI chatbots in clinical research.

Materials and Methods

Study Design

A list of randomized controlled trials (RCTs) in obstetrics and maternal–fetal medicine (MFM) published in 2020 by top medical and obstetrics and gynecology (OBGYN) journals was obtained from a previous publication in the American Journal of Obstetrics and Gynecology Maternal-Fetal Medicine (AJOG MFM).[16] The top 10 general medical journals publishing original research in OBGYN, based on the 2019 Journal Citation Report by Clarivate Analytics, were chosen. Eligible RCTs were published in journals with the highest impact factors (IFs), including the British Medical Journal, the New England Journal of Medicine, the Journal of the American Medical Association, and the Lancet. OBGYN journals that published original research in the field of obstetrics and MFM with the 10 highest IFs were also selected. These journals include the AJOG, Ultrasound in Obstetrics and Gynecology, Obstetrics and Gynecology, and the British Journal of Obstetrics and Gynecology. Doulaveris et al[16] included AJOG MFM even though it had not yet been assigned an IF, as it published its first issue in 2019 but had a strong focus on publishing RCTs. See ►Table 1 for a complete list of the included RCTs.

Three AI large language models (LLMs) were selected for evaluation based on their popularity and performance in natural language processing tasks, as well as their availability to the public at no cost. The models selected for testing were ChatGPT (OpenAI, San Francisco, CA), Google Bard (Google, Mountain View, CA), and Chatsonic (Writesonic, San Francisco, CA). ChatGPT, which initially popularized AI for scientific writing and research, was chosen alongside Google Bard and Chatsonic, listed among the best ChatGPT alternatives. The language understanding of both ChatGPT and Chatsonic is based on the Generative Pre-trained Transformer (GPT) technology, whereas Google Bard uses Language Model for Dialogue Applications (LaMDA). However, Google Bard and Chatsonic rely on the internet to provide real-time information for their interactions with users, whereas ChatGPT's knowledge is limited to its training cutoff in September 2021. During the study, GPT 3.5 was available to the public for free, whereas GPT 4.0 was only available through paid subscriptions. ChatGPT (GPT 3.5) was queried from late March to early April 2023, whereas Google Bard and Chatsonic (GPT 3.5) were queried in late April 2023.

Data Collection

Three authors used the chosen AI chatbots to find references related to the selected RCTs. The AI chatbots were prompted to describe an RCT published in 2020 that addressed the same objective as described in the RCT's abstract on PubMed. The AI chatbot was prompted to provide the reference of the study in the American Medical Association (AMA) Manual of Style. The prompt format used for the LLMs was standardized across all authors. If the AI chat tool failed to provide the desired reference or presented a legitimate article on the same topic but not the desired one, the authors would repeat the prompt, specifying the name of the journal where the target RCT was published. In cases where multiple responses were received from the AI chatbot, the first response was used for accuracy assessment. A comprehensive search of

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Table 1 Selected obstetric randomized controlled trials published in 2020

Journal name	Month and year of publication	Publication title
NEJM	July 2020	Respiratory syncytial virus vaccination during pregnancy and effects in infants[18]
NEJM	December 2020	Antenatal dexamethasone for early preterm birth in low-resource countries[19]
JAMA	June 2020	Effect of a patient-centered decision support tool on rates of trial of labor after previous cesarean delivery: the PROCEED randomized clinical trial[20]
JAMA	September 2020	Effect of prophylactic negative pressure wound therapy vs standard wound dressing on surgical-site infection in obese women after cesarean delivery[21]
The Lancet	January 2020	Low-dose aspirin for the prevention of preterm delivery in nulliparous women with a singleton pregnancy (ASPIRIN): a randomized, double-blind, placebo-controlled trial[22]
AJOG	March 2020	MAVRIC: a multicenter randomized controlled trial of transabdominal vs transvaginal cervical cerclage[23]
AJOG	April 2020	Enhanced recovery after surgery at cesarean delivery to reduce postoperative length of stay: a randomized controlled trial[24]
AJOG	May 2020	Nonopioid versus opioid analgesia after hospital discharge following cesarean delivery: a randomized equivalence trial[25]
AJOG	May 2020	Early gestational diabetes screening in obese women: a randomized controlled trial[26]
AJOG	July 2020	Study to reduce infection prior to elective cesarean deliveries (STRIPES): a randomized clinical trial of chlorhexidine[27]
AJOG	July 2020	Perioperative prophylactic internal iliac artery balloon occlusion in the prevention of postpartum hemorrhage in placenta previa: a randomized controlled trial[28]
AJOG	August 2020	Evaluation of an external fetal electrocardiogram monitoring system: a randomized controlled trial[29]
AJOG	September 2020	Prelabor screening at term using the cerebroplacental ratio and placental growth factor: a pragmatic randomized open-label phase 2 trial[30]
AJOG	November 2020	Prospective, randomized, double-blind, placebo-controlled evaluation of the pharmacokinetics, safety and efficacy of recombinant antithrombin versus placebo in preterm preeclampsia[31]
AJOG	December 2020	Physical examination–indicated cerclage in twin pregnancy: a randomized controlled trial[32]
UOG	January 2020	Elective delivery at 34 weeks vs routine obstetric care in fetal gastroschisis: randomized controlled trial[33]
UOG	April 2020	Multicenter randomized trial exploring effects of simulation-based ultrasound training on obstetricians' diagnostic accuracy: value for experienced operators[34]
OG	February 2020	Ropivacaine and ketorolac wound infusion for post-cesarean delivery analgesia: a randomized controlled trial[35]
OG	April 2020	Cephalic elevation device for second-stage cesarean delivery: a randomized controlled trial[36]
OG	May 2020	Internal iliac artery balloon occlusion for placenta previa and suspected placenta accreta: a randomized controlled trial[37]
OG	May 2020	Uterine exteriorization compared with in situ repair of hysterotomy after cesarean delivery: a randomized controlled trial[38]
OG	May 2020	Double-balloon device for 6 compared with 12 hours for cervical ripening: a randomized controlled trial[39]
OG	June 2020	Effect of digital step counter feedback on mobility after cesarean delivery: a randomized controlled trial[40]
OG	September 2020	Reducing the risk for postpartum depression in adolescent mothers: a randomized controlled trial[41]
OG	September 2020	Outpatient Foley catheter for induction of labor in nulliparous women: a randomized controlled trial[42]

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Table 1 (*Continued*)

Journal name	Month and year of publication	Publication title
OG	November 2020	Foley bulb added to an oral misoprostol induction protocol: a cluster randomized trial[43]
OG	December 2020	Alternate dosing protocol for magnesium sulfate in obese women with preeclampsia: a randomized controlled trial[44]
BJOG	January 2020	Effects of oral probiotic supplements on vaginal microbiota during pregnancy: a randomized, double-blind, placebo-controlled trial with microbiome analysis[45]
BJOG	February 2020	Restrictive versus routine episiotomy among Southeast Asian term pregnancies: a multicentre randomized controlled trial[46]
BJOG	March 2020	Pravastatin for early-onset pre-eclampsia: a randomized, blinded, placebo-controlled trial[47]
BJOG	March 2020	Effect of couple-based cognitive behavioral intervention on prevention of postnatal depression: multisite randomized controlled trial[48]
BJOG	April 2020	Induction of labor using prostaglandin E2 as an inpatient versus balloon catheter as an outpatient: a multicentre randomized controlled trial[49]
BJOG	June 2020	Mindfetalness to increase women's awareness of fetal movements and pregnancy outcomes: a cluster-randomized controlled trial including 39 865 women[50]
BJOG	June 2020	STRAWB2 (Stress and Wellbeing After Childbirth): a randomized controlled trial of targeted self-help materials to prevent post-traumatic stress disorder following childbirth[51]
BJOG	September 2020	Antenatal magnesium sulfate for the prevention of cerebral palsy in infants born preterm: a double-blind, randomized, placebo-controlled, multi-center trial[52]
BJOG	September 2020	A randomized trial of caesarean wound coverage: exposed versus dressed[53]
BJOG	October 2020	Twelve-hour fasting compared with expedited oral intake in the initial inpatient management of hyperemesis gravidarum: a randomized trial[54]
BJOG	December 2020	Vaginal compared with intramuscular progestogen for preventing preterm birth in high-risk pregnant women (VICTORIA study): a multicentre, open-label randomized trial and meta-analysis[55]
BJOG	December 2020	Intrauterine versus external tocodynamometry in monitoring labor: a randomized controlled clinical trial[56]
AJOG MFM	May 2020	The effect of intrauterine resuscitation by maternal hyperoxygenation on perinatal and maternal outcome: a randomized controlled trial[57]
AJOG MFM	August 2020	A randomized trial of the bactericidal effects of chlorhexidine vs povidone-iodine vaginal preparation[58]
AJOG MFM	August 2020	A randomized controlled trial of liposomal bupivacaine for pain following obstetrical laceration[59]
AJOG MFM	August 2020	Impact of a novel smartphone application on low-income, first-time mothers' breastfeeding rates: a randomized controlled trial[60]
AJOG MFM	August 2020	A randomized trial of dapagliflozin and metformin, alone and combined, in overweight women after gestational diabetes mellitus[61]

Abbreviations: AJOG, American Journal of Obstetrics and Gynecology; MFM, Maternal-Fetal Medicine; BJOG, British Journal of Obstetrics and Gynecology; JAMA, Journal of American Medical Association; NEJM, New England Journal of Medicine; OG, Obstetrics & Gynecology; UOG, Ultrasound in Obstetrics and Gynecology.

PubMed and Google Scholar was conducted to thoroughly analyze each citation's accuracy.

To ensure relevant data collection, a standardized data form was used to extract information from PubMed and the AI chatbots. The data of interest included the study's title, its citation from PubMed and screenshots of the conversation with the LLMs regarding each study. Three authors (E.M.G., J.A.M., A.B.D.) utilized a standardized data extraction form on Microsoft Word. Additionally, two authors (E.M.G., J.A.M.) independently utilized a standardized data spreadsheet on Microsoft Excel to track the correctness of the references provided by the chatbots, which was cross-checked for accuracy. Disputes were resolved through consensus during discussions among the authors. If consensus could not be reached, the senior author (L.S.R.) was consulted for resolution.

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This study was not supported by any funding agency in the commercial, not-for-profit, or public sectors. The study was deemed exempt from the institutional review board approval as all studies examined included completely decoded and publicly available data.

Outcomes

The accuracy of each AI model in correctly citing references was measured as the primary evaluation metric. Each reference was carefully analyzed for the correctness of authors, publication title, journal name, and digital object identifier (DOI). The assessment of citation accuracy involved checking all required components according to the AMA Manual of Style guidelines, including authors, publication title, journal name, and DOI. A citation was considered completely accurate if it correctly listed all the components of the reference. Additionally, any citations that were partially correct were noted, along with which components of the AMA reference were accurately cited. Additionally, we assessed adherence to the AMA Manual of Style guidelines by evaluating whether the listed journal names aligned with the NLM abbreviations, if provided by the AI chatbots.

Statistical Analysis

To compare the performance of the AI models, a permutation test was performed. For each permutation, the accuracy of the AI models in each category was randomly shuffled across the entire data collection to create a permuted dataset. Next, the accuracy of each model was calculated using the permuted dataset. The process of shuffling and calculating accuracy was repeated 10,000 times to generate a null distribution of accuracies under the assumption of no difference in performance between the models. The observed accuracy of each model was compared with the null distribution to determine if the difference in accuracy between the models was statistically significant. The p-value represents the probability of observing a difference between the two AI tools' performances as extreme as, or more extreme than, the observed difference, assuming the null hypothesis is true. A significance level of $p < 0.05$ was used to determine statistical significance. If the p-value is below 0.05, the null hypothesis is rejected and concludes that there is evidence to suggest that the two AI tools have different performances in that category. On the other hand, if the p-value is greater than the significance level, we fail to reject the null hypothesis and conclude that there is not enough evidence to suggest that the two AI tools have different performances in that category. Statistical analysis and permutation tests were performed using R Studios version 4.3 (Poist, Boston, MA).

Results

Of the 44 obstetric randomized trials published in the selected medical and OBGYN journals during 2020, the AI chatbots were only able to cite 7 references completely correct (5.3%; 7 out of 132 searched). ChatGPT was able to cite one randomized trial correctly (2.4%; 1/44), Google Bard cited 6 correctly (13.6%; 6/44), and Chatsonic cited all requested RCTs incorrectly. Additionally, the authors were not able to use a universal prompt for all three AI LLMs. The authors discovered that the prompt used for ChatGPT (the first LLM to be prompted by the authors) had to be modified for Google Bard and Chatsonic to respond to the prompt. Once a prompt was found to generate a response from a particular chatbot, that specific prompt was universally adopted and used by all authors for interactions with that specific chatbot. See ►Fig. 1 for an example of a fabricated reference provided by ChatGPT for a randomized trial published by AJOG MFM and see ►Fig. 2 for an example of Google Bard correctly citing a source.

Google Bard was most consistently able to cite accurate information in relation to the desired RCTs. Google Bard significantly outperformed Chatsonic in every reference component category and outperformed ChatGPT in all but one reference component. When citing the correct journal name there was no significant difference in performance between ChatGPT and Google Bard ($p = 0.053$). There was also not a significant difference between the performance of ChatGPT and Chatsonic in any studied category. See ►Table 2 for a summary of the permutation analysis and comparison of each AI model performance.

Google Bard was the most successful at providing the correct DOI, 38.6% (17 out of 44 studies). The majority of references from all AI models studied were noted to provide DOIs for unrelated studies or even DOIs that do not exist. Interestingly, in 35 out of 44 studies Chatsonic provided a hyperlink to the desired study, despite being unable to cite it correctly. See ►Fig. 3 for an example of Chatsonic providing a fictitious reference with a hyperlink to the desired RCT's PubMed page.

Discussion

Principal Findings

The analysis of the responses provided by ChatGPT, Google Bard, and Chatsonic revealed that AI chatbots have varying accuracy in referencing scientific articles. Frequently, the studied AI tools combined segments of keywords from different sources, resulting in citations that do not exist in scientific literature. All three AI models were more likely to generate fictitious references and present them as authentic citations rather than notifying the user that they could not provide a reliable reference due to limitations in their programming. Despite Chatsonic's ability to access the internet for up-to-date information and utilizing the same framework as ChatGPT, it performed the lowest in providing accurate references. However, Chatsonic was often observed to correctly summarize the RCT of interest and provide a hyperlink to the relevant PubMed page. Among the studied chatbots, Google Bard demonstrated the highest reliability in scientific writing outperforming the popular ChatGPT.

Results in the Context of What is Known

This study contributes to the growing body of literature on the limitations of AI chatbots in producing accurate citations

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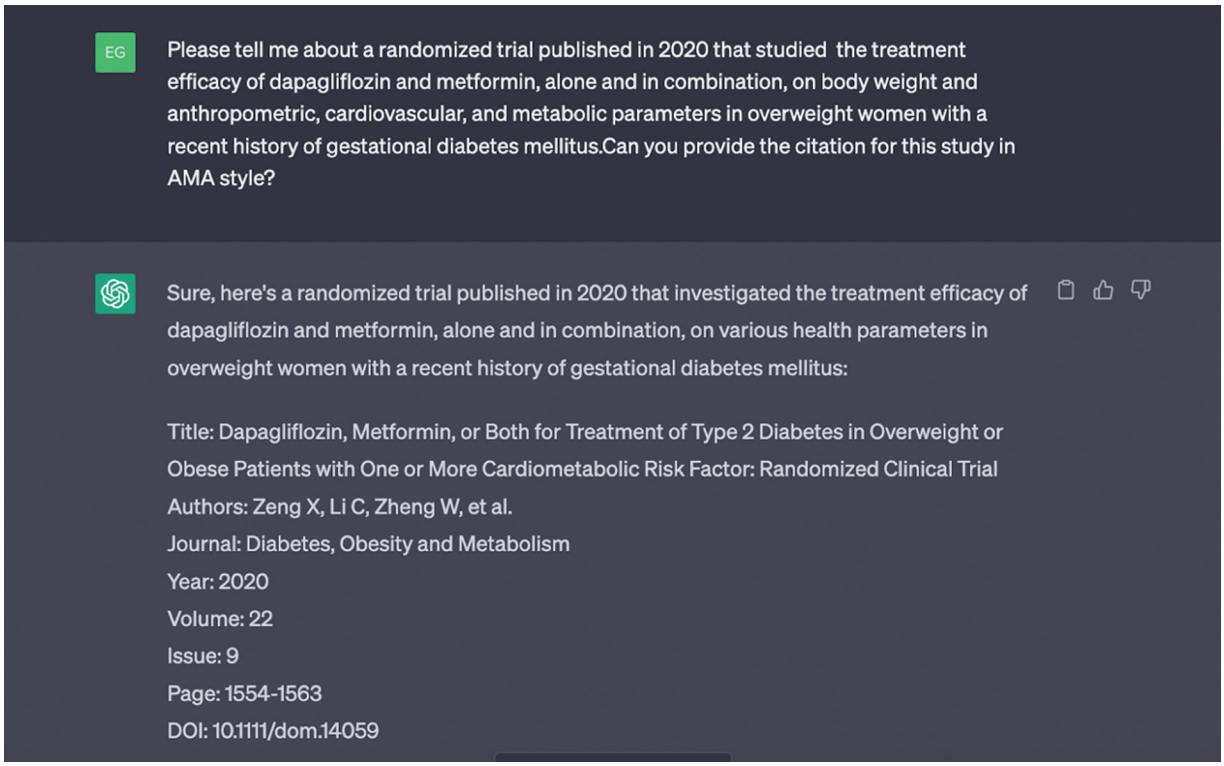


Fig. 1 Example of a fictitious reference by ChatGPT. DOI provided is for an unrelated article. Reference for randomized study being requested: Elkind-Hirsch et al 2020.[61]

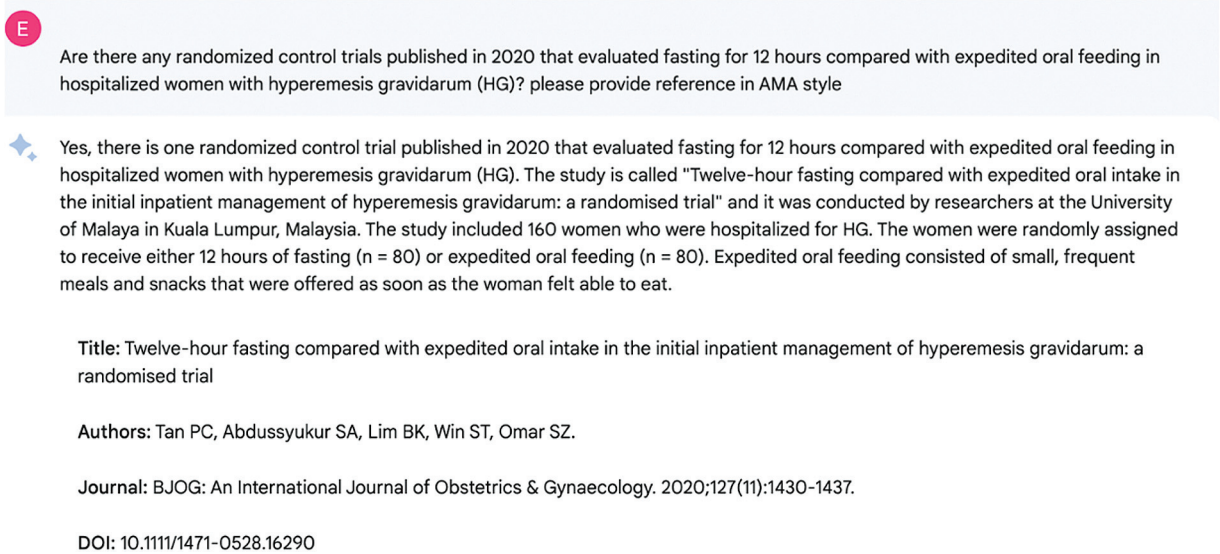


Fig. 2 Example of an accurate reference provided by Google Bard. Reference for randomized study being requested: Tan et al 2020.[54]

and their tendency to generate fictitious references. While previous research has focused primarily on ChatGPT,[6–8] particularly in relation to artificial hallucinations or confabulations,[9–15] this study expands upon the topic by comparing the reliability of ChatGPT to other popular AI chatbots in providing citations to scientific articles. Furthermore, this study employed advanced statistical analysis to demonstrate that the performance differences among the evaluated chatbots is significant.

Research and Clinical Implications

While AI chatbots can be helpful in generating ideas and writing assistance, they cannot be considered dependable sources for generating accurate references. With the increasing use of AI in scientific writing, it may be necessary for journals to mandate authors to disclose the use of AI in manuscript preparation.[17] In addition, authors must be cautious when utilizing AI for scientific writing and literature search to ensure the reliability of information being disseminated to the

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Table 2 Summary of permutation analysis

	ChatGPT vs. Google Bard			Chatsonic vs. Google Bard		
	ChatGPT	Google Bard	*p*-Value	Chatsonic	Google Bard	*p*-Value
Correct lead author (%)	6.8	31.8	0.007	3.5	31.8	<0.001
Correct secondary authors (%)	4.5	22.7	0.038	4.5	22.7	0.022
Correct publication title (%)	18.2	68.2	<0.001	15.9	68.2	<0.001
Correct journal name (%)	40.9	63.6	0.053	29.5	63.6	0.006
Correct DOI (%)	2.3	38.6	<0.001	0	38.6	<0.001
Partially correct reference (%)[a]	52.3	84.1	0.002	31.8	84.1	<0.001
Completely correct reference (%)	2.3	13.6	0.122	0	13.6	0.030
	ChatGPT vs. Chatsonic					
	ChatGPT	Chatsonic	*p*-Value			
Correct lead author (%)	6.8	3.5	0.626			
Correct secondary authors (%)	4.5	4.5	1.000			
Correct publication title (%)	18.2	15.9	1.000			
Correct journal name (%)	40.9	29.5	0.358			
Correct DOI (%)	2.3	0	1.000			
Partially correct reference (%)[a]	52.3	31.8	0.063			
Completely correct reference (%)	2.3	0	1.000			

Abbreviation: DOI, digital object identifier.
[a]If any component of the reference was correct.

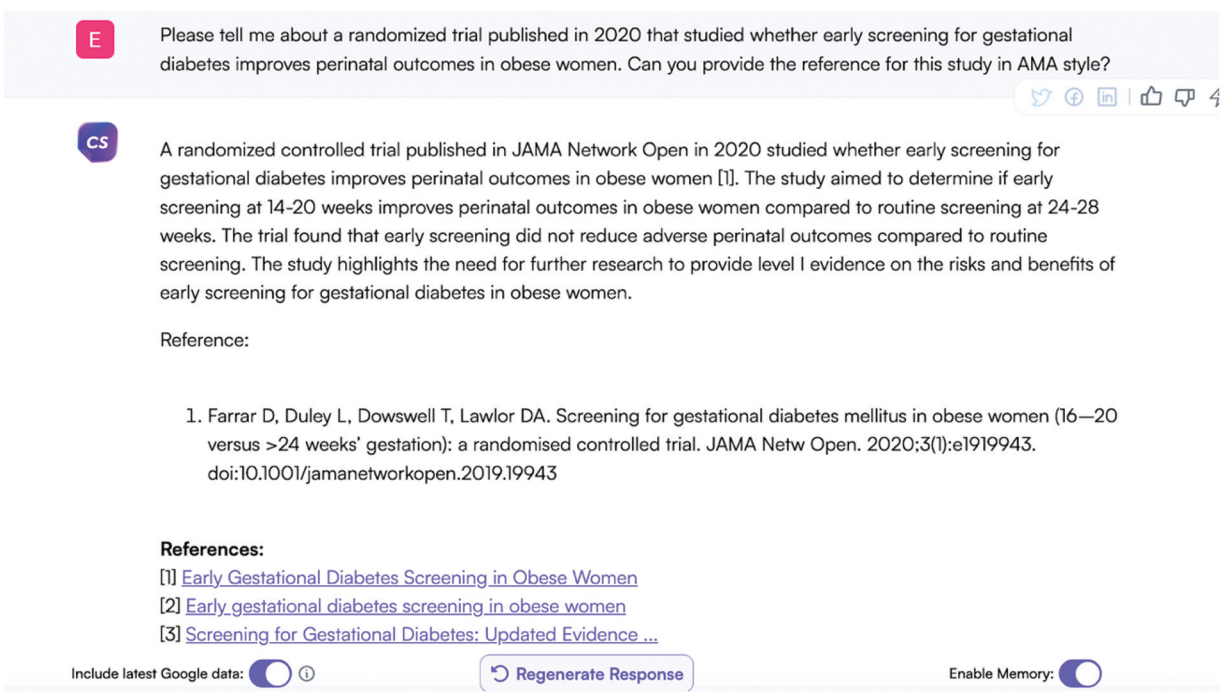


Fig. 3 Example of a fictitious reference by Chatsonic. The description of the RCT is accurate and the first hyperlink is the PubMed abstract page for the requested RCT. However, the reference is completely fictitious and the DOI does not exist. Reference for randomized study being requested: Harper et al 2020.[26]

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scientific community. Similarly, health care providers and clinicians must exercise caution while using AI chatbots to answer clinical questions to avoid recommending potentially ineffective or harmful treatments.

Strengths and Limitations

This study demonstrates several key strengths that contribute to its validity and reliability. First, it offers originality by comparing three prominent AI models, each with distinct focuses and intended applications. Second, to ensure consistency in the data collection process, the chatbots were prompted in a stepwise manner, following a predetermined protocol.

Third, the use of a permutation test for statistical analysis is a significant strength of the study. The permutation test is a robust nonparametric method that allows for accurate comparisons between the performances of different AI tools in multiple categories, which helps provide a reliable assessment of the differences in accuracy. By employing a standardized evaluation metric, the accuracy of each model is quantitatively measured, providing a clear and comparable assessment of their performance.

Although this cross-sectional study has notable strengths, it is important to acknowledge its limitations when interpreting the findings. First, the study only evaluated three AI models, which may not represent the full capabilities of all AI chatbots available.

Second, the findings and conclusions drawn from the study may have limited generalizability beyond the specific dataset utilized. The performance of the AI models in citing references can vary depending on various dataset characteristics, such as the field of study, the language used, or specific citation styles employed. Therefore, caution should be exercised when applying the study's findings to different datasets or domains.

Third, the outputs of the AI models varied significantly based on the wording of the prompt and the time at which the prompt was given. There were rare instances where each AI model declined to provide a reference, citing an inability to perform the requested task. However, it is worth noting that the AI models would attempt to generate a reference when posed with the same prompt at a different time or by a different author. This variability in responses is an inherent characteristic of AI systems based on natural language processing, which may affect the replicability of the study.

Fourth, the reproducibility of this study is limited by the rapidly evolving nature of AI technology. At the time of this study, the authors only had access to GPT 3.5, whereas newer versions such as GPT 4.0 may provide different results.

Conclusion

The use of AI in scientific writing and literature searching has both strengths and limitations that require careful consideration. Advanced algorithms enable rapid scanning of vast amounts of scientific literature, making AI an efficient tool for information searching, data analysis, and automated content generation. Additionally, AI systems can uncover patterns, correlations, and trends that may not be readily apparent to human researchers, further facilitating data exploration. These tools make AI invaluable in assisting researchers in the writing process. However, it is crucial to acknowledge that AI is imperfect and has noteworthy constraints.

Accurate referencing is a fundamental aspect of scientific writing that cannot be overlooked. Authors must ensure that their references are relevant, accurate, and consistent to enhance their research credibility and foster progress in their respective fields. The results of this study highlight the importance of continuing to use traditional platforms such as Google Scholar and PubMed for scientific literature searches, rather than solely relying on the information provided by AI.

AI holds significant promise in revolutionizing the process of scientific writing and, by recognizing and mitigating these limitations, clinicians and researchers can harness the strengths of AI while ensuring the responsible and ethical use of these technologies. Ultimately, collaborative partnerships between AI systems and researchers can lead to synergistic advancements in medical literature and scientific progress.

Conflict of Interest
None declared.

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